SUPERSTORE SALES PERFORMANCE ANALYSIS

A stakeholder-ready report for Alfido Tech

Project objective
Evaluate sales performance across time, regions, product categories and products; identify seasonality and translate the findings into practical sales actions.

DATASET SNAPSHOT	VALUE
Rows after cleaning	9,800
Unique orders	4,922
Unique customers	793
Sales period	03 Jan 2015 – 30 Dec 2018
Total recorded sales	$2,261,536.78

Prepared using the supplied Superstore CSV dataset. All currency values are reported in the dataset’s apparent US-dollar convention.

1. Executive Summary

The dataset contains **9,800 cleaned transaction lines** across **4,922 distinct orders** and **793 customers**, spanning **03 Jan 2015 – 30 Dec 2018**. Total recorded sales were **$2,261,536.78**, with average order value of **$459.48**. Technology was the largest category by sales, and the West was the strongest region by sales revenue. November was the strongest calendar month when all years are combined.

KPI	RESULT	INTERPRETATION
Total sales	$2,261,536.78	Sum of Sales field
Unique orders	4,922	Distinct Order_ID values
Average order value	$459.48	Total sales divided by unique orders
Unique customers	793	Distinct Customer_ID values
Average sales per customer	$2,851.87	Descriptive average; not customer lifetime value

Important: this file does not contain profit, cost, discount, product quantity, website sessions or conversion events. Profit margin and conversion rate therefore cannot be calculated from this dataset. Sales revenue is not the same as profit.

Main takeaways

• **Category:** Technology contributed 36.6% of sales and led the three categories.

• **Region:** West contributed 31.4% of total sales; South had the lowest regional sales.

• **Seasonality:** November was the highest-sales calendar month overall, followed by December and September.

• **Product concentration:** the leading product by revenue was Canon imageCLASS 2200 Advanced Copier ($61,599.82).

2. Data Cleaning & Methodology

The original file contained 9,800 rows and 18 columns. The analysis retained rows with valid order ID, order date, sales, region, category and product name. Exact duplicate rows were checked and none were found. Eleven postal-code values were missing; postal code was not required for the requested sales analysis. Date fields were parsed into date types and year/month fields were derived for trend analysis.

CHECK	RESULT / TREATMENT
Source rows	9,800
Exact duplicate rows	0 found; no duplicate rows removed
Missing postal codes	11 missing values; retained because not essential to sales summaries
Critical fields	Order_ID, Order_Date, Sales, Region, Category and Product_Name checked
Date coverage	03 Jan 2015 – 30 Dec 2018
Sales data type	Converted to numeric; rows with invalid critical fields excluded
Analysis grain	Line-level sales records; order-level metrics use distinct Order_ID

KPI definitions

• **Total sales / revenue:** sum of the Sales column.

• **Average order value (AOV):** total sales divided by the number of distinct orders.

• **Average sales per customer:** total sales divided by distinct customer IDs; descriptive only.

• **Profit margin:** requires profit and revenue (profit ÷ revenue); profit is absent.

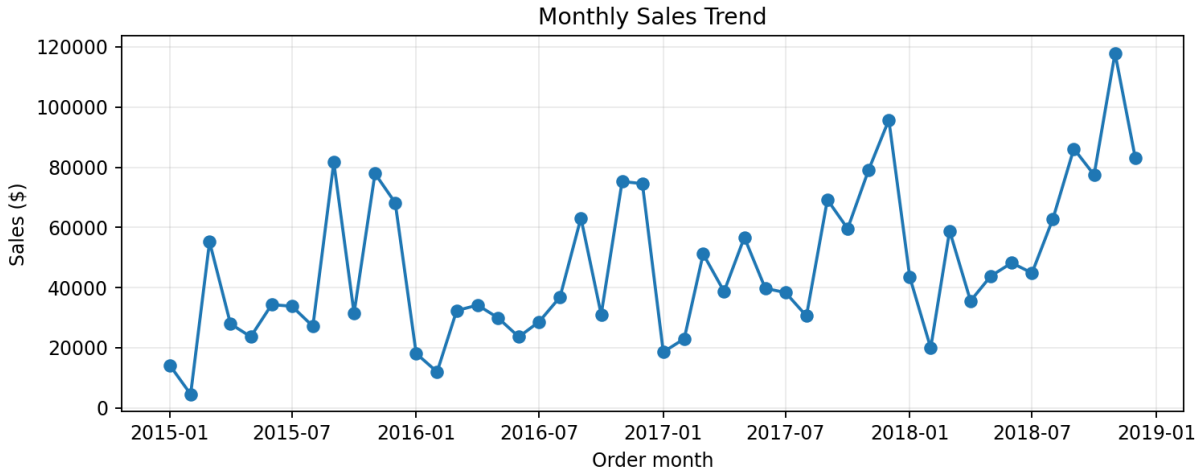
• **Conversion rate:** requires conversions and a traffic/session denominator; neither is present.

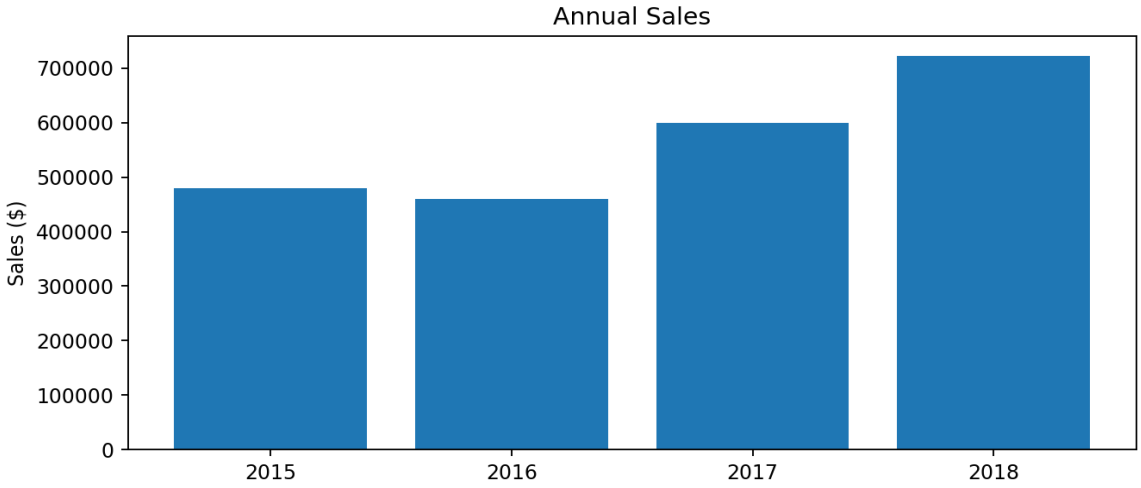
Interpretation note

These findings describe historical sales, not causal effects. Regional revenue differences can reflect customer mix, order frequency, product availability or other factors. The dataset’s date strings are interpreted day-first for consistency with the source file’s mixed date formatting. Validate date conventions with the data owner before using the report for formal forecasting.

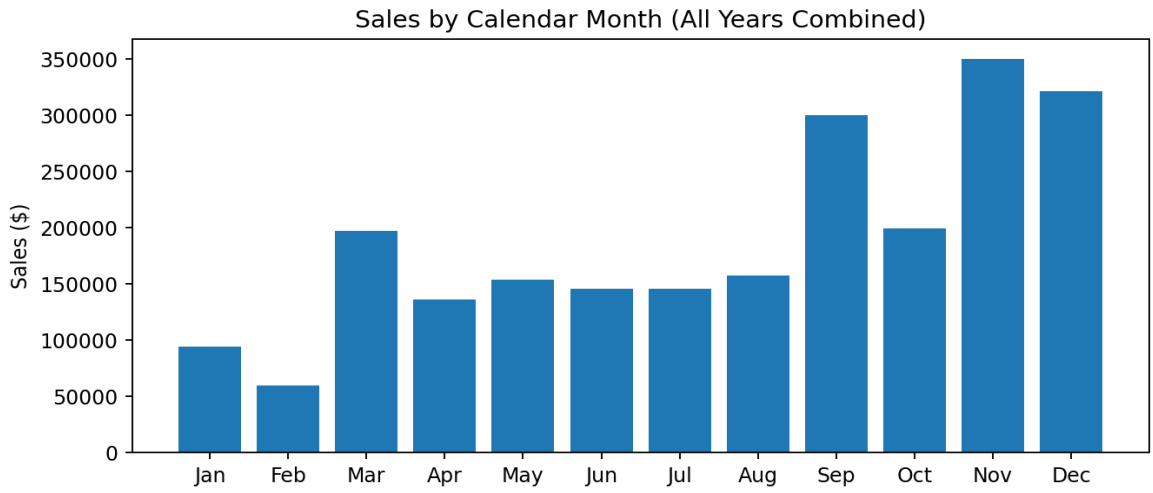
3. Sales Trends & Seasonality

Monthly sales vary over the four-year period. The monthly line chart is useful for spotting peaks and troughs, but month-to-month comparisons should be interpreted alongside promotions, stock availability and business changes that are not captured in the file.





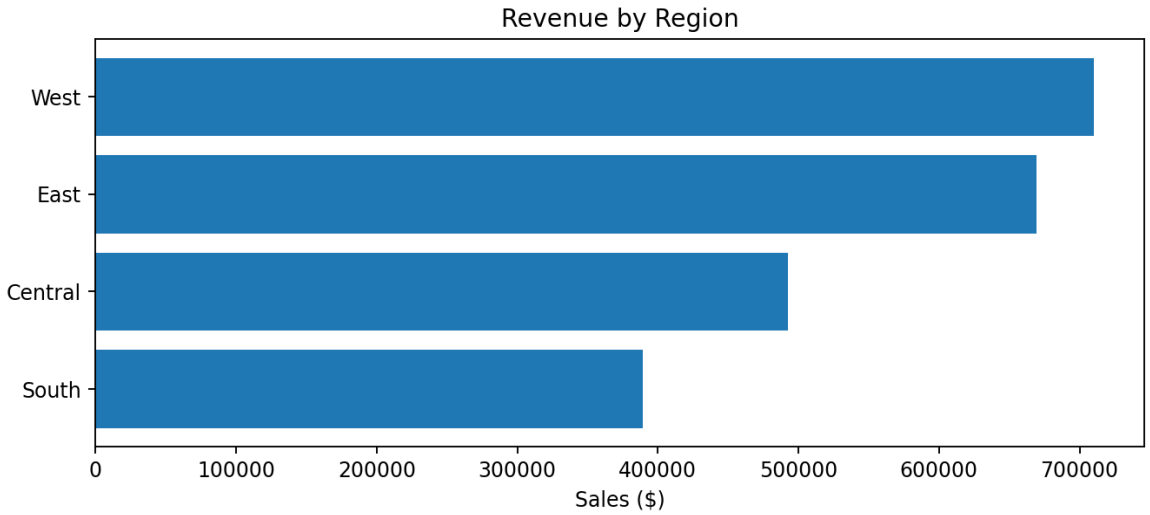
Seasonality across calendar months



November was the highest-sales calendar month at $350,161.71; Nov sales were about 15.5% of all recorded sales. February was the lowest at $59,371.12. These are aggregated values across all years, not a forecast for the next year.

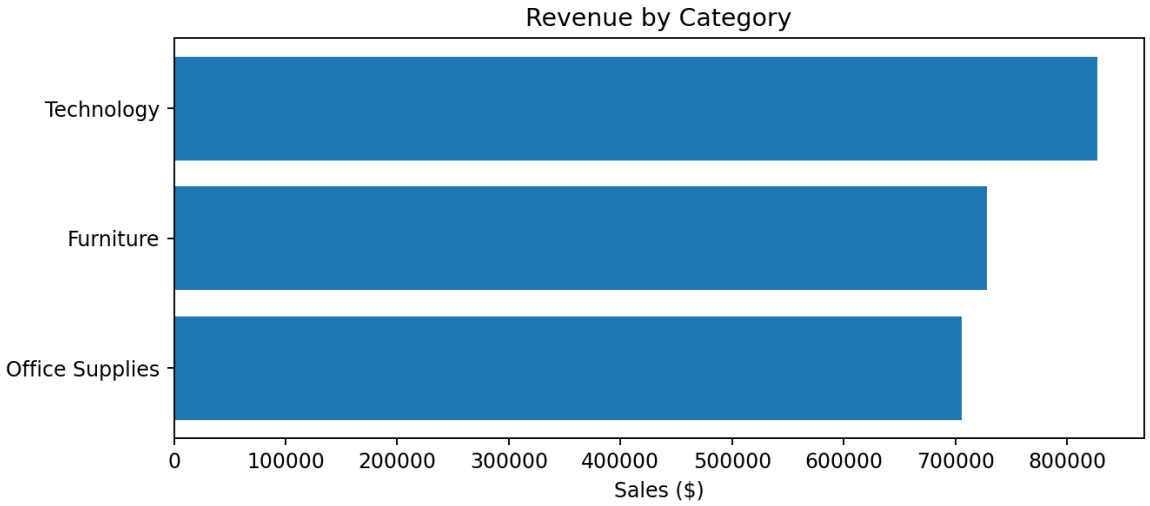
4. Regional & Category Performance

Regional performance



REGION	SALES	SHARE OF SALES
West	$710,219.68	31.4%
East	$669,518.73	29.6%
Central	$492,646.91	21.8%
South	$389,151.46	17.2%

Category performance

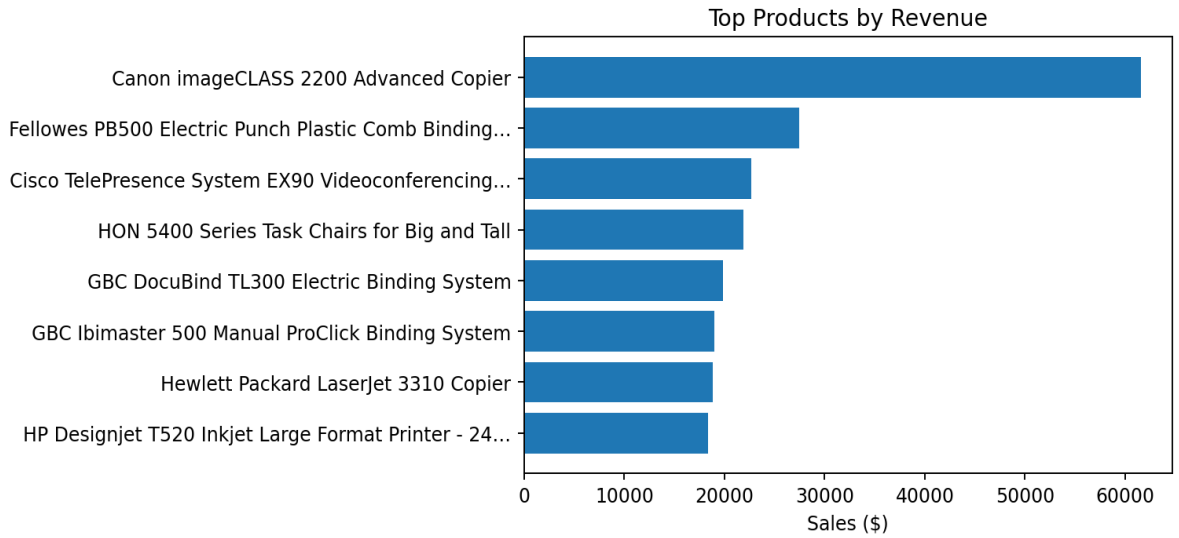


CATEGORY	SALES	SHARE OF SALES
Technology	$827,455.87	36.6%
Furniture	$728,658.58	32.2%
Office Supplies	$705,422.33	31.2%

West is the highest-sales region and South the lowest-sales region. Technology leads category sales. Treat these as prioritization signals rather than proof of profitability: no profit or cost field is available.

5. Product Performance

Top products by recorded sales

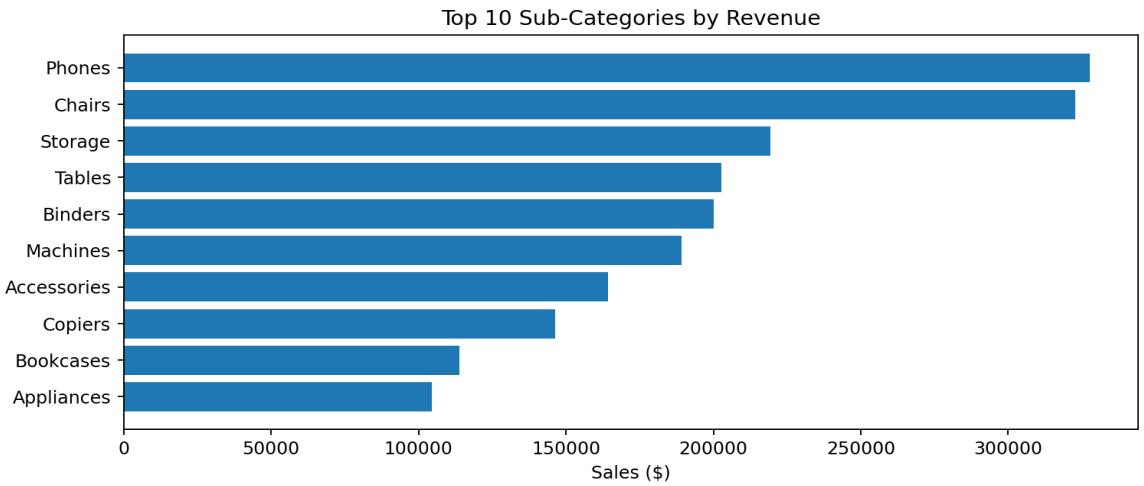


RANK	PRODUCT	SALES
1	Canon imageCLASS 2200 Advanced Copier	$61,599.82
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	$27,453.38
3	Cisco TelePresence System EX90 Videoconferencing Unit	$22,638.48
4	HON 5400 Series Task Chairs for Big and Tall	$21,870.58
5	GBC DocuBind TL300 Electric Binding System	$19,823.48

Lowest products by recorded sales (among products in this file)

PRODUCT	SALES
Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac	$1.62
Avery 5	$5.76
Xerox 20	$6.48
Grip Seal Envelopes	$7.07
Acme Serrated Blade Letter Opener	$7.63
Avery Hi-Liter Pen Style Six-Color Fluorescent Set	$7.70
Avery Hi-Liter Comfort Grip Fluorescent Highlighter, Yellow Ink	$7.80
Xerox 1989	$7.97

The top product by revenue is Canon imageCLASS 2200 Advanced Copier. The lowest-revenue products shown above may simply have sold in small quantities; low revenue alone does not prove weak demand or poor profitability. Before discontinuing products, review units sold, margins, inventory costs, returns and strategic importance.


Top 10 Sub-Categories by Revenue
Phones
Chairs
Storage
Tables
Binders
Machines
Accessories
Copiers
Bookcases
Appliances
0
50000
100000
150000
200000
250000
300000
Sales ($)

6. Five Tactical Improvements for Alfido Tech

1. Plan around seasonal demand

Build a monthly demand calendar. Start inventory and campaign planning ahead of the observed November–December peak; test targeted offers during weaker months such as February. Compare results year-over-year before locking budgets.

2. Protect and promote leading products

Maintain availability and strong product-page visibility for top revenue contributors, including the Canon imageCLASS 2200 Advanced Copier. Check gross margin, stock-outs, returns and supplier lead times before increasing promotion.

3. Run region-specific growth experiments

Use West and East as reference markets for successful assortment and merchandising practices. Test small, measurable campaigns in South, the lowest-sales region, and compare incremental revenue, order count and contribution margin against a control where possible.

4. Improve basket size with relevant bundles

Test complementary bundles in Technology and Furniture. Measure AOV, attach rate, conversion and margin per order. Avoid bundling low-margin items simply to lift revenue.

5. Build a stronger performance measurement system

Add product cost, profit, discount, quantity, returns, inventory and marketing spend to the dataset. If online funnel data exists, add visits/sessions and conversions. Create a weekly dashboard for revenue, order count, AOV, margin, stock-outs and campaign ROI.

Suggested 30–60–90 day action plan

WHEN	ACTION	SUCCESS MEASURE
First 30 days	Validate date conventions; establish baseline dashboard; identify stock-out risks and data gaps.	Trusted baseline; availability KPI definitions.
Days 31–60	Pilot seasonal campaign, regional test and 1–2 product bundles.	Incremental sales, AOV, margin, campaign cost
Days 61–90	Compare test results; scale winning activities; stop or revise weak tests.	Positive incremental contribution; fewer stock-outs

7. Dashboard Blueprint & Limitations

Recommended dashboard layout

SECTION	VISUAL / FILTERS
KPI strip	Total sales, unique orders, AOV, customers; add profit margin only after profit/cost data is supplied
Trend	Monthly sales line chart with year and date filters.
Regional	Sales by region; add state/city drill-down.
Product & category	Category/sub-category bar charts; top and bottom products by sales.
Seasonality	Sales by calendar month, with year selector to compare seasons.
Filters	Order date, region, category, sub-category, segment, ship mode.

Data limitations and responsible use

• **No profit or cost fields:** profitability, gross margin and contribution cannot be evaluated.

• **No quantity field:** product unit volumes and average selling price cannot be calculated.

• **No web traffic/conversion events:** conversion rate cannot be calculated.

• **No discounts or returns:** discount effectiveness and return rates cannot be assessed.

• **Historical data only:** the data ends in 2018; verify current sales and operating context before applying decisions.

• **Low-revenue products:** rankings are based on total sales value only and should not be used alone for delisting decisions.

Conclusion

The dataset provides a clear baseline for revenue analysis: total sales are $2,261,536.78, the West region leads sales, Technology is the largest category, and November is the strongest aggregated calendar month. Alfido Tech should use these patterns to guide controlled tests in seasonal planning, product availability, regional campaigns and basket-building. The next priority is to enrich the data with profit, cost, quantity and funnel metrics so that future decisions optimize profitable growth rather than revenue alone.

Appendix: reproducibility

Source: user-supplied Superstore sales CSV extracted from archive.zip. Cleaning and calculations were performed with Python and pandas; charts were generated with Matplotlib and the report assembled with ReportLab. Currency formatting follows the apparent USD convention in the dataset. Key metrics use distinct Order_ID and Customer_ID counts; average order value is calculated from order-level sales totals.